

Reproducing and Extending [“TradingAgents: Multi-Agents LLM Financial Trading Framework”](#)

Project Overview: TradingAgents Framework

This project reproduces and extends the findings of *TradingAgents: Multi-Agents LLM Financial Trading Framework*. The original paper proposes a collaborative financial trading architecture where multiple Large Language Model (LLM) agents assume specialized roles similar to a professional trading firm (e.g., fundamental, sentiment, and technical analysis, trade execution, and risk management).

The Research Gap

While the authors demonstrate that their collaborative architecture outperforms traditional rule-based strategies, their experimental design leaves a critical question unresolved.

- The evaluation only compares the full system against traditional rule-based strategies.
- The experiments do not isolate the causal contributions of specific components like role specialization, debate, and risk-management.
- It remains unclear whether performance gains stem from the multi-agent organization itself or simply from increased information processing.

Phase 1: Baseline Reproduction

We will first verify the paper's core claims using the authors' open-source implementation to ensure the published results can be replicated.

- **Method:** Run the TradingAgents framework on historical stock data using the original backtesting procedure.
- **Output:** Daily buy, sell, or hold decisions generated through coordinated agent reasoning.
- **Evaluation Metrics:** Cumulative return, annualized return, Sharpe ratio, and maximum drawdown.

Phase 2: Organizational Ablation Study

To test whether improvements arise from organizational structure rather than the mere use of LLMs, we will analyze how trading performance changes as the complexity of the agent society increases. We will construct four progressively richer variants of the system while keeping the dataset and evaluation pipeline unchanged.

System Configurations:

1. **Single Generalist:** One agent processes all available information sources and directly generates trading recommendations.
2. **Specialized Analysts:** Multiple agents separately analyze different signal types, but without inter-agent debate.
3. **Research Debate:** Adds a bullish-bearish debate stage allowing agents with opposing perspectives to evaluate analyst reports.
4. **Full Architecture:** Replicates the complete original framework, including the risk-management team that reviews and adjusts final decisions.

Phase 3: Information Source Extension

We will conduct a small follow-up experiment examining the impact of different data inputs.

- **Method:** Selectively remove specific inputs, such as news articles or social media sentiment signals.
- **Goal:** Provide a clearer understanding of which information sources contribute most to improved financial decision-making.

Expected Contribution

Together, these experiments will isolate exactly which components of multi-agent LLM systems drive performance, providing a rigorous analysis while remaining feasible within the computational limits of a course project.